Tutorial 3

Z-transform Foundations

SC3102: Signals, Systems, and Transforms

Question 1

Definition and Re-weighting

Recall: DTFT and Absolute Summability

- The DTFT of a discrete-time signal $x[n]$ is

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-\mathrm{j}\Omega n}.$$

- Absolute summability guarantees convergence:

$$\sum_{n} |x[n]| < \infty.$$

- However, not guaranteed for **many** important signals

Recall: Z-transform and ROC

- The Z-transform reweights the input signal and then computes the DTFT:

$$X(z = re^{j\Omega}) = \sum_{n=-\infty}^{\infty} \underbrace{x[n]r^{-n}}_{\text{Re-weighted Signal}} e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

- Its **region of convergence (ROC)** was z such that:

$$\sum_{n=-\infty}^{\infty} |x[n]|\,|z|^{-n} = \sum_{n=-\infty}^{\infty} |x[n]|\,r^{-n} < \infty.$$

- A Z-transform is the expression **and** its ROC

Question 1(a): Definition and the Re-weighted Signal

- Define the Z-transform of $x[n]$
- Suppose $z = re^{j\Omega}$ lies in its ROC, with $r > 0$. Which signal $w[n]$ has DTFT

$$W(\Omega) = X(re^{j\Omega})?$$

- Express $x[n]$ in terms of $w[n]$.

1(a): The Re-weighted Signal w

- Let $z = re^{j\Omega}$ lie in the ROC, with $r > 0$.

$$X(re^{j\Omega}) = \sum_n \underbrace{r^{-n}x[n]}_{w[n]} e^{-j\Omega n}.$$

- From the expression, $W(\Omega)$ is the DTFT of:

$$\boxed{w[n] = r^{-n}x[n]}, \qquad \boxed{x[n] = r^n w[n]}.$$

Recall: Exponential DTFT

- For a complex number q with $|q| < 1$,

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}.$$

- Taking $q = ae^{-j\Omega}$ gives the DTFT pair:

$$a^n u[n] \longleftrightarrow \frac{1}{1-ae^{-j\Omega}}, \qquad |a| < 1.$$

Question 1(b): Recovering the Original Signal

- We know the circle $r = |z| = 3$ lies in the ROC and

$$X(3e^{\mathrm{j}\Omega}) = \frac{1}{1 - \frac{\mathrm{j}}{2}e^{-\mathrm{j}\Omega}}.$$

- **Goal:** Recover $x[n]$.

1(b): Inverse DTFT Pair and Rescaling

- On the circle $r = 3$, we know

$$w[n] = 3^{-n}x[n] \qquad \text{and} \qquad W(\Omega) = X(3e^{\mathrm{j}\Omega}) = \frac{1}{1 - \frac{\mathrm{j}}{2}e^{-\mathrm{j}\Omega}}.$$

- Remember the inverse pair:

$$a^n u[n] \longleftrightarrow \frac{1}{1 - ae^{-\mathrm{j}\Omega}}, \qquad |a| < 1.$$

- Matching we get $a = \mathrm{j}/2$:

$$w[n] = 3^{-n}x[n] = \left(\frac{\mathrm{j}}{2}\right)^n u[n] \qquad \Longrightarrow \qquad \boxed{x[n] = \left(\frac{3\mathrm{j}}{2}\right)^n u[n].}$$

Question 1(c–d): Re-weighting Signals

- **(c)** Find the $r > 0$ for which $r^{-n}x[n]$ has a DTFT:

$$x_R[n] = (2\mathrm{j})^n u[n], \qquad x_L[n] = 2^n u[-n-1].$$

- **(d)** Do these signals have nonempty ROCs?

$$g[n] = 2^n, \qquad h[n] = 2^{|n|}.$$

1(c): The Right-sided Signal

- The weighted right-sided signal, for $r > 0$, is

$$w_R[n] = r^{-n} x_R[n] = r^{-n}(2\mathrm{j})^n u[n] = \left(\frac{2\mathrm{j}}{r}\right)^n u[n].$$

- Absolute summability requires

$$\begin{aligned}\sum_n |w_R[n]| &= \sum_{n=0}^{\infty} |2\mathrm{j}|^n r^{-n} = \sum_{n=0}^{\infty} 2^n r^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{2}{r}\right)^n < \infty \quad \Longleftrightarrow \quad \frac{2}{r} < 1 \quad \Longleftrightarrow \quad \boxed{r > 2}.\end{aligned}$$

1(c): The Left-sided Signal

- For the left-sided signal $x_L[n] = 2^n u[-n-1]$,

$$\sum_n |r^{-n} x_L[n]| = \sum_{n=-\infty}^{-1} |2^n| r^{-n} = \sum_{n=-\infty}^{-1} \left(\frac{2}{r}\right)^n.$$

- The negative-time sum uses $m = -n = 1, 2, \ldots$:

$$\sum_{n=-\infty}^{-1} \left(\frac{2}{r}\right)^n = \sum_{m=1}^{\infty} \left(\frac{2}{r}\right)^{-m} = \sum_{m=1}^{\infty} \left(\frac{r}{2}\right)^m < \infty$$

$$\iff \quad \frac{r}{2} < 1 \quad \iff \quad \boxed{0 < r < 2}.$$

1(d): The Exponential Signal $g[n] = 2^n$

- First, split the signal into left and right parts:

$$g[n] = 2^n\Big(u[n] + u[-n-1]\Big) = \underbrace{2^n u[n]}_{g_R[n]} + \underbrace{2^n u[-n-1]}_{g_L[n]}.$$

- Both parts must converge for the same radius:

$$\begin{aligned}\sum_n |g[n]| r^{-n} &= \sum_{n=0}^{\infty} 2^n r^{-n} + \sum_{n=-\infty}^{-1} 2^n r^{-n} \\ &= \sum_n |g_R[n]| r^{-n} + \sum_n |g_L[n]| r^{-n}.\end{aligned}$$

1(d): Convergence of the Two Parts of $g[n]$

- The right-sided part is $g_R[n] = 2^n u[n]$:

$$\sum_n |g_R[n]| r^{-n} = \sum_{n=0}^{\infty} 2^n r^{-n} = \sum_{n=0}^{\infty} \left(\frac{2}{r}\right)^n < \infty$$

$$\iff \quad \frac{2}{r} < 1 \quad \iff \quad \boxed{r > 2}.$$

- The left-sided part is $g_L[n] = 2^n u[-n-1]$; $m = -n$:

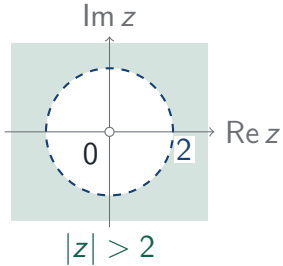
$$\sum_{n=-\infty}^{-1} 2^n r^{-n} = \sum_{m=1}^{\infty} 2^{-m} r^m = \sum_{m=1}^{\infty} \left(\frac{r}{2}\right)^m < \infty$$

$$\iff \quad \frac{r}{2} < 1 \quad \iff \quad \boxed{0 < r < 2}.$$

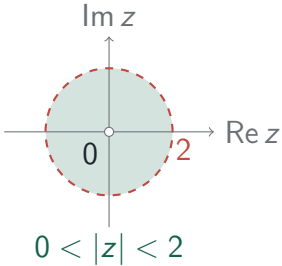
1(d): The Two Regions for $g[n]$

- Do **both** left and right parts converge?

$$g_R[n] = 2^n u[n]$$



$$g_L[n] = 2^n u[-n-1]$$



- There is no value of r and the ROC is empty!

1(d): The Two-sided Signal $h[n] = 2^{|n|}$

- First, observe that this signal can be written as follows:

$$h[n] = 2^{|n|} = \begin{cases} 2^n, & n \geq 0, \\ 2^{-n}, & n \leq -1. \end{cases}$$

- We can split the signal as before and evaluate absolute summability:

$$h[n] = \underbrace{2^n u[n]}_{h_R[n]} + \underbrace{2^{-n} u[-n-1]}_{h_L[n]}.$$

$$\sum_n |h[n]| r^{-n} = \sum_{n=0}^{\infty} 2^n r^{-n} + \sum_{n=-\infty}^{-1} 2^{-n} r^{-n}.$$

1(d): Convergence of the Two Parts of $h[n]$

- The right-sided part is $h_R[n] = 2^n u[n]$:

$$\sum_n |h_R[n]| r^{-n} = \sum_{n=0}^{\infty} 2^n r^{-n} = \sum_{n=0}^{\infty} \left(\frac{2}{r}\right)^n < \infty$$

$$\iff \quad \frac{2}{r} < 1 \quad \iff \quad \boxed{r > 2}.$$

- The left-sided part is $h_L[n] = 2^{-n} u[-n-1]$; $m = -n$:

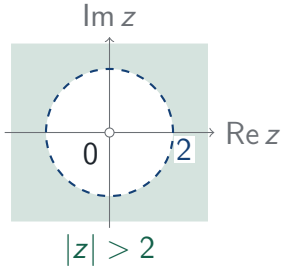
$$\sum_{n=-\infty}^{-1} 2^{-n} r^{-n} = \sum_{m=1}^{\infty} 2^m r^m = \sum_{m=1}^{\infty} (2r)^m < \infty$$

$$\iff \quad 2r < 1 \quad \iff \quad \boxed{0 < r < \frac{1}{2}}.$$

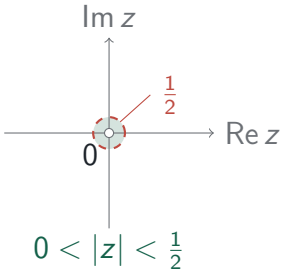
1(d): The Two Regions for $h[n]$

- Do both parts converge?

$$h_R[n] = 2^n u[n]$$



$$h_L[n] = 2^{-n} u[-n-1]$$



- Again, the ROC is empty!

Question 2

Transforms, ROCs, and the DTFT

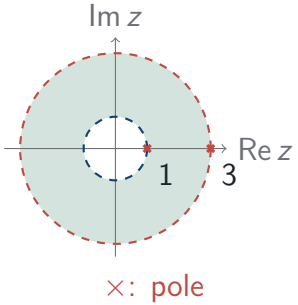
Recall: Poles Delineate Possible ROCs

- After cancelling common factors, the **poles** are the roots of the denominator
- The circles containing the poles separate possible ROCs

$$X(z) = \frac{z+3}{(z-1)(z-3)}$$

- The possible ROCs are

$$0 < |z| < 1, \quad 1 < |z| < 3, \quad |z| > 3.$$



2(a): Z-transform of the Finite Signal

- Finite signal: $x_1[n] = \delta[n] - 3\delta[n-1] + 2\delta[n-2]$.

$$\delta[n-k] \longleftrightarrow z^{-k}$$

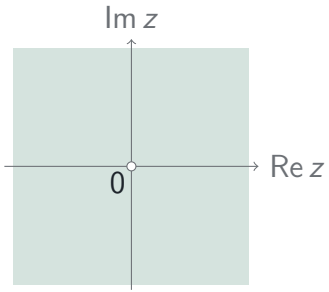
- Taking the Z-transform:

$$\begin{aligned} X_1(z) &= 1 - 3z^{-1} + 2z^{-2} \\ &= \frac{z^2}{z^2} - \frac{3z}{z^2} + \frac{2}{z^2} = \frac{z^2 - 3z + 2}{z^2} = \boxed{\frac{(z-1)(z-2)}{z^2}}. \end{aligned}$$

2(a): ROC of the Finite Signal

- The signal is finite! Its ROC is the entire Z-plane except $z = 0$.
- Absolute convergence:

$$\sum_n |x_1[n]| r^{-n} = 1 + 3r^{-1} + 2r^{-2} < \infty \qquad (r > 0).$$



- Region of convergence:

$$\boxed{0 < |z| < \infty.}$$

2(a): Z-transform of the Ramp Signal

- The given signal is the standard ramp minus a unit step:

$$x_2[n] = nu[n] = \Big((n+1) - 1\Big)u[n] = (n+1)u[n] - u[n].$$

- The standard ramp is the convolution of two unit steps:

$$(u * u)[n] = \sum_{m=-\infty}^{\infty} u[m]u[n-m] = (n+1)u[n].$$

- Since $U(z) = z/(z-1)$, convolution and linearity give with ROC $|z| > 1$:

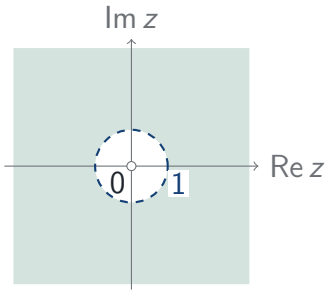
$$X_2(z) = U(z)^2 - U(z) = \frac{z^2}{(z-1)^2} - \frac{z}{z-1} = \boxed{\frac{z}{(z-1)^2}}.$$

2(a): ROC of the Ramp Signal

- Both terms have the ROC $|z| > 1$.

- $X_2(z)$ has a (double) pole at $z = 1$
- We get the **exterior** ROC:

$$\boxed{|z| > 1.}$$



2(a): Z-transform of the Left-sided Signal

- Left-sided exponential: $x_3[n] = (-2)^n u[-n-1]$; $m = -n \geq 1$.

$$X_3(z) = \sum_{n=-\infty}^{-1} (-2)^n z^{-n} = \sum_{m=1}^{\infty} (-2)^{-m} z^m = \sum_{m=1}^{\infty} \left(-\frac{z}{2}\right)^m.$$

- The above is a geometric series:

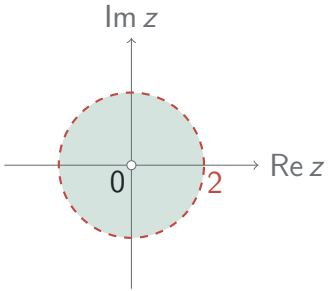
$$X_3(z) = \frac{-z/2}{1-(-z/2)} = \frac{-z/2}{(z+2)/2} = \boxed{-\frac{z}{z+2}}.$$

when $|z| < 2$.

2(a): ROC of the Left-sided Signal

- Absolute convergence:

$$\sum_{n=-\infty}^{-1} |(-2)^n| r^{-n} = \sum_{m=1}^{\infty} 2^{-m} r^m$$
$$= \sum_{m=1}^{\infty} \left(\frac{r}{2}\right)^m < \infty.$$



- Region of convergence:

$$\frac{r}{2} < 1 \iff r < 2, \qquad \boxed{0 < |z| < 2.}$$

2(a): Z-transform of the Two-sided Signal

- Again, split the signal into two:

$$x_{4R}[n] = \left(\frac{1}{2}\right)^n u[n], \qquad x_{4L}[n] = -3^n u[-n-1].$$

- Exponential Z-transform pairs:

$$p^n u[n] \leftrightarrow \frac{z}{z-p}, \text{ ROC: } |z| > |p|, \quad -p^n u[-n-1] \leftrightarrow \frac{z}{z-p}, \text{ ROC: } |z| < |p|$$

- Putting both these together:

$$X_4(z) = X_{4R}(z) + X_{4L}(z) = \boxed{\frac{z}{z-\frac{1}{2}} + \frac{z}{z-3}} \qquad \text{ROC: } \frac{1}{2} < |z| < 3.$$

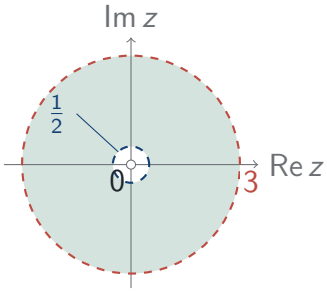
2(a): ROC of the Two-sided Signal

- Right tail, $n \geq 0$:

$$\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n r^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2r}\right)^n < \infty$$
$$\iff r > \frac{1}{2}.$$

- Left tail, $m = -n \geq 1$:

$$\sum_{n=-\infty}^{-1} 3^n r^{-n} = \sum_{m=1}^{\infty} \left(\frac{r}{3}\right)^m < \infty$$
$$\iff r < 3.$$

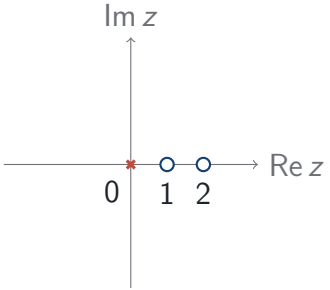


2(b): Other Possible ROCs for X_1

- The Z-transform is:

$$X_1(z) = \frac{(z-1)(z-2)}{z^2}$$

- The ROC is $0 < |z| < \infty \implies$ No other ROC!

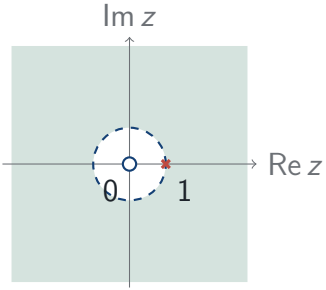


2(b): Possible ROCs for X_2

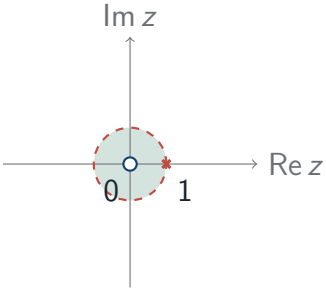
- The Z-transform is:

$$X_2(z) = \frac{z}{(z-1)^2}$$

From (a): $|z| > 1$



Other ROC: $0 < |z| < 1$

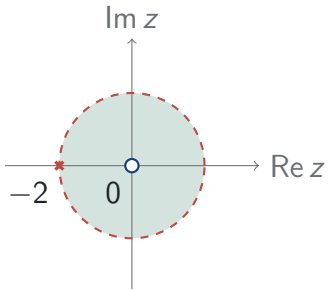


2(b): Possible ROCs for X_3

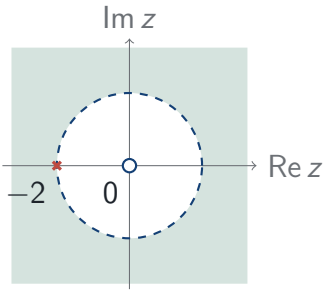
- The Z-transform is:

$$X_3(z) = -\frac{z}{z+2}$$

From (a): $0 < |z| < 2$



Other ROC: $|z| > 2$

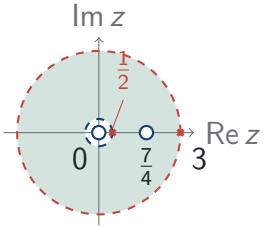


2(b): Possible ROCs for X_4

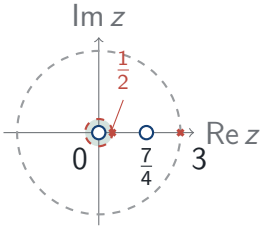
- The Z-transform is:

$$X_4(z) = \frac{2z(z - \frac{7}{4})}{(z - \frac{1}{2})(z - 3)}$$

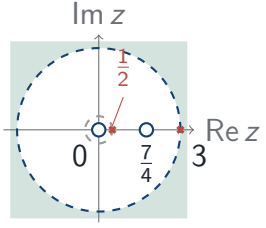
From (a): $\frac{1}{2} < |z| < 3$



Other ROC: $0 < |z| < \frac{1}{2}$



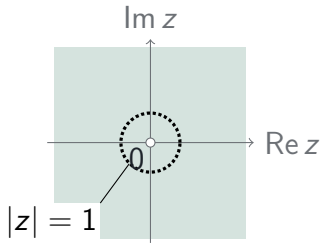
Other ROC: $|z| > 3$



2(c): DTFTs of the Finite Signal and Ramp

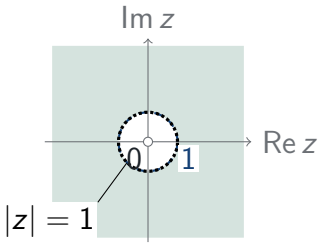
- The DTFT exists when the ROC contains the unit circle $|z| = 1$.

$$X_1(z) = \frac{(z-1)(z-2)}{z^2}$$



DTFT exists.

$$X_2(z) = \frac{z}{(z-1)^2}$$

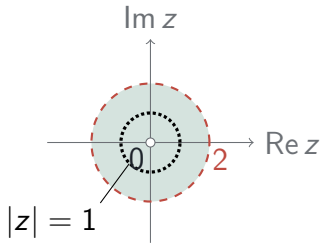


No DTFT.

2(c): DTFTs of the Left- and Two-sided Signals

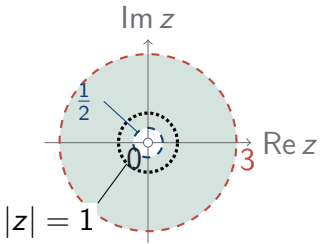
- The original ROCs of $x_3[n]$ and $x_4[n]$ both contain $|z| = 1$.

$$X_3(z) = -\frac{z}{z+2}$$



DTFT exists.

$$X_4(z) = \frac{2z(z-\frac{7}{4})}{(z-\frac{1}{2})(z-3)}$$



DTFT exists.

Question 3

Inversion Methods

Recall: Partial-Fraction Method

- Any rational expression has a partial-fraction decomposition:

$$X(z) = \frac{B(z)}{A(z)} = P(z) + \sum_{i=1}^{K} \sum_{k=1}^{m_i} \frac{c_{i,k}}{(z - p_i)^k}.$$

- The p_i are the poles, and m_i their multiplicities. $P(z)$ is the polynomial part.
- Use known transform pairs and the ROC to invert each term.
- Typically, use exponential pairs in this course

Recall: Time Delay and Partial Fractions

- A delay by k samples contributes the factor z^{-k}, for integer $k \geq 0$:

$$s[n] \longleftrightarrow S(z) \quad \Longrightarrow \quad s[n-k] \longleftrightarrow z^{-k}S(z).$$

- With $k = 1$, we get terms in the partial fraction expansion:

$$p^{n-1}u[n-1] \longleftrightarrow z^{-1}\frac{z}{z-p} = \frac{1}{z-p}, \qquad |z| > |p|.$$

Recall: Long Division Method

- Repeatedly divide numerator by denominator
- The **ROC** determines the direction of the division:

Exterior ROC:	Descending powers of z,
Interior ROC:	Increasing powers of z.

- The coefficient of z^{-n} is $x[n]$:

$$X(z) = \sum_n x[n]z^{-n} = \cdots + x[-1]z + x[0] + x[1]z^{-1} + \cdots .$$

3(a): Sinusoidal Inverse

- The known cosine pair is

$$\rho^n \cos(\omega_0 n) u[n] \quad \longleftrightarrow \quad \frac{z(z - \rho\cos\omega_0)}{z^2 - 2\rho z\cos\omega_0 + \rho^2}, \qquad |z| > \rho.$$

- The given transform matches $\rho = 1/2$, $\omega_0 = \pi/2$:

$$X(z) = \frac{z^2}{z^2 + 1/4} = \frac{z(z - \frac{1}{2}\cos\frac{\pi}{2})}{z^2 - z\cos\frac{\pi}{2} + \frac{1}{4}}, \qquad |z| > 1/2.$$

- The cosine pair therefore gives:

$$\boxed{x[n] = \left(\frac{1}{2}\right)^n \cos\left(\frac{\pi n}{2}\right) u[n].}$$

3(b): The Partial-Fraction Inverse

- We are given the Z-transform:

$$A(z) = \frac{3z-7}{z^2-5z+6} = \frac{3z-7}{(z-2)(z-3)}, \qquad |z| > 3.$$

- The partial-fraction form is

$$A(z) = \frac{C}{z-2} + \frac{D}{z-3}.$$

- **Hint:** Multiply both sides by $z-p$ and substitute $z=p$.

3(b): Identifying C and D

- Let's first multiply by $z-2$:

$$(z-2)A(z) = \frac{(z-2)(3z-7)}{(z-2)(z-3)} = \frac{3z-7}{z-3} = C + D\frac{z-2}{z-3}$$

- After substituting $z=2$, the D term disappears on the right!

$$\frac{3(2)-7}{2-3} = \frac{-1}{-1} = C + D\frac{0}{-1} = C \quad \Longrightarrow \quad \boxed{C=1}$$

3(b): Identifying C and D

- Similarly multiply by $z-3$:

$$(z-3)A(z) = \frac{(z-3)(3z-7)}{(z-2)(z-3)} = \frac{3z-7}{z-2} = C\frac{z-3}{z-2} + D$$

- After substituting $z=3$, the C term disappears instead

$$\frac{3(3)-7}{3-2} = 2 = C\frac{0}{1} + D = D \quad \Longrightarrow \quad \boxed{D=2}$$

3(b): Inverting the Z-transform

- The supplied ROC $|z| > 3$ is the exterior ROC of the Z-transform:

$$A(z) = \frac{1}{z-2} + \frac{2}{z-3}.$$

- These correspond to *delayed* exponential:

$$\frac{1}{z-2} \quad \longleftrightarrow \quad 2^{n-1}u[n-1],$$
$$\frac{2}{z-3} = 2\frac{1}{z-3} \quad \longleftrightarrow \quad 2 \cdot 3^{n-1}u[n-1].$$

- The inverse is the sum:

$$\boxed{a[n] = \left(2^{n-1} + 2 \cdot 3^{n-1}\right)u[n-1].}$$

3(c): The Long Division Method

- We are required to find $y[1], y[2], y[3]$ from the Z-transform

$$Y(z) = \frac{2 + z^{-1}}{1 - \frac{1}{2}z^{-1}}, \qquad |z| > 1/2.$$

- We write it out in rational form below:

$$Y(z) = \frac{z(2 + z^{-1})}{z(1 - \frac{1}{2}z^{-1})} = \frac{2z + 1}{z - \frac{1}{2}}.$$

- We are given the **exterior** ROC

3(c): Long Division

- We eliminate remainder terms with leading denominator term

$$
\begin{array}{r|l}
 & 2 + 2z^{-1} + z^{-2} + \frac{1}{2}z^{-3} + \cdots \\
\hline
z - \frac{1}{2} & 2z + 1 \\
- & 2z - 1 \\
\hline
 & 2 \\
 - & 2 - z^{-1} \\
\hline
 & z^{-1} \\
 - & z^{-1} - \frac{1}{2}z^{-2} \\
\hline
 & \frac{1}{2}z^{-2} \\
 - & \frac{1}{2}z^{-2} - \frac{1}{4}z^{-3} \\
\hline
 & \frac{1}{4}z^{-3} \\
 & \vdots
\end{array}
$$

3(c): Reading the Signal Values

- The quotient by long division reads:

$$Y(z) = 2 + 2z^{-1} + z^{-2} + \frac{1}{2}z^{-3} + \cdots .$$

- The coefficients of z^{-1}, z^{-2}, z^{-3} give

$$\boxed{y[1] = 2, \qquad y[2] = 1, \qquad y[3] = \frac{1}{2}.}$$

3(d): Reading Series

- The given series is:

$$V(z) = \cdots + \frac{1}{8}z^3 + \frac{1}{4}z^2 + \frac{1}{2}z + 1 + \frac{1}{3}z^{-1} + \frac{1}{9}z^{-2} + \cdots .$$

- The coefficient of z^{-n} is $v[n]$:

$$\boxed{v[-2] = \frac{1}{4}, \quad v[-1] = \frac{1}{2}, \quad v[0] = 1, \quad v[1] = \frac{1}{3}, \quad v[2] = \frac{1}{9}.}$$

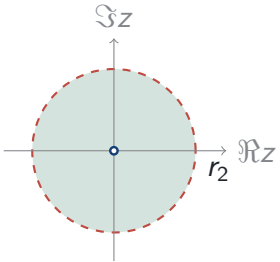
Question 4

Selecting the ROC and the Signal

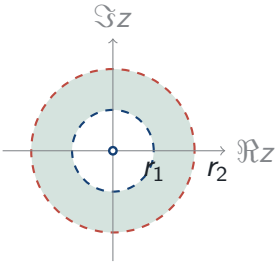
Recall: Possible ROC Shapes

- For a rational Z-transform, pole circles bound the ROC and are excluded

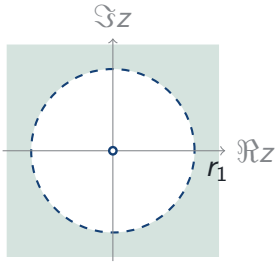
Interior: $0 < |z| < r_2$



Annular: $r_1 < |z| < r_2$



Exterior: $|z| > r_1$



Question 4(a–b): Poles, ROCs, and Inverses

- A discrete-time signal $f[n]$ has Z-transform expression:

$$F(z) = \frac{2z^2 + 2z}{z^2 + 2z - 8}.$$

- **(a)** Plot its poles and zeros and give the possible ROCs.
- **(b)** For $r = 1, 3, 5$, find the ROC and $f[n]$ if the circle $|z| = r$ lies in the ROC.

4(a): Computing Poles and Zeros

- We can factorize the expression to get:

$$F(z) = \frac{2z(z+1)}{(z-2)(z+4)}.$$

- Poles:

$$z - 2 = 0 \Rightarrow z = 2, \qquad z + 4 = 0 \Rightarrow z = -4.$$

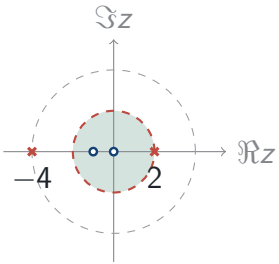
- Zeros:

$$z = 0 \qquad \text{or} \qquad z + 1 = 0 \Rightarrow z = -1.$$

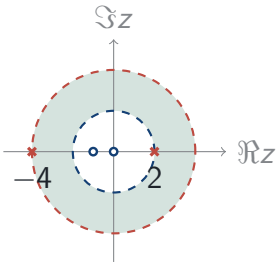
4(a): Possible ROCs

- The pole radii are $|2| = 2$ and $|-4| = 4$.

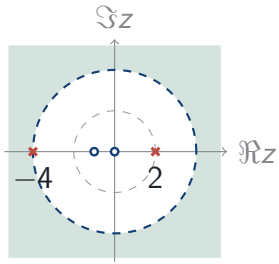
$0 < |z| < 2$



$2 < |z| < 4$



$|z| > 4$



4(b): Inversion with Partial Fractions

- For this, it will be convenient to use different partial fractions:

$$F(z) = \frac{2z^2 + 2z}{(z-2)(z+4)} = z\left(\frac{2z+2}{(z-2)(z+4)}\right) = C\frac{z}{z-2} + D\frac{z}{z+4}.$$

- Let's first multiply by $z-2$:

$$(z-2)F(z) = \frac{(z-2)(2z^2+2z)}{(z-2)(z+4)} = \frac{2z^2+2z}{z+4} = Cz + D\frac{z(z-2)}{z+4}.$$

- After substituting $z = 2$, the D term disappears:

$$\frac{2(2)^2 + 2(2)}{2+4} = 2 = 2C \quad \Longrightarrow \quad \boxed{C = 1}.$$

4(b): Inversion with Partial Fractions

- Similarly multiply by $z+4$:

$$(z+4)F(z) = \frac{(z+4)(2z^2+2z)}{(z-2)(z+4)} = \frac{2z^2+2z}{z-2} = C\frac{z(z+4)}{z-2} + Dz.$$

- After substituting $z=-4$, the C term disappears:

$$\frac{2(-4)^2+2(-4)}{-4-2} = -4 = -4D \quad \Longrightarrow \quad \boxed{D=1}.$$

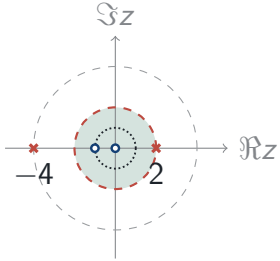
- Therefore, we can decompose F as:

$$\boxed{F(z) = \frac{z}{z-2} + \frac{z}{z+4}}.$$

4(b): Inverting the circle $r = 1$

- Since $0 < r < 2$, this is the interior ROC: $\boxed{0 < |z| < 2}$.

$$\frac{z}{z-2} \longleftrightarrow -2^n u[-n-1],$$
$$\frac{z}{z-(-4)} \longleftrightarrow -(-4)^n u[-n-1].$$



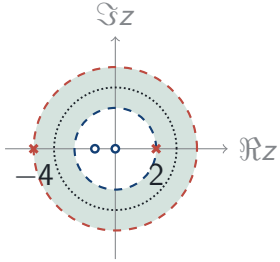
- Take left-sided inverse for both:

$$\begin{aligned} f[n] &= -2^n u[-n-1] - (-4)^n u[-n-1] \\ &= \boxed{-\left(2^n + (-4)^n\right) u[-n-1]}. \end{aligned}$$

4(b): Inverting the circle $r = 3$

- Since $2 < r < 4$, this is the annular ROC: $\boxed{2 < |z| < 4}$.

$$\frac{z}{z-2} \longleftrightarrow 2^n u[n],$$
$$\frac{z}{z-(-4)} \longleftrightarrow -(-4)^n u[-n-1].$$



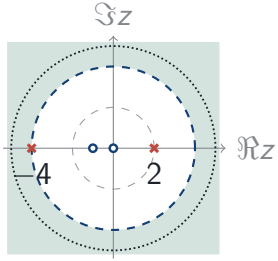
- Take right-sided for the first term and left-sided for the second:

$$\boxed{f[n] = 2^n u[n] - (-4)^n u[-n-1].}$$

4(b): Inverting the circle $r = 5$

- Since $r > 4$, this is the exterior ROC: $\boxed{|z| > 4}$.

$$\frac{z}{z-2} \longleftrightarrow 2^n u[n],$$
$$\frac{z}{z-(-4)} \longleftrightarrow (-4)^n u[n].$$



- Take the right-sided inverse for both:

$$\begin{aligned} f[n] &= 2^n u[n] + (-4)^n u[n] \\ &= \boxed{\left(2^n + (-4)^n\right) u[n]}. \end{aligned}$$

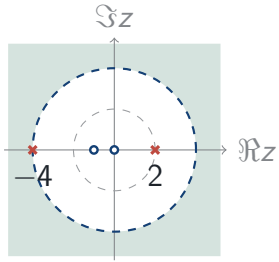
Question 4(c–e): Inverting Based on Signal Constraints

- For the same F:
- **(c)** Which ROC applies if $f[n] = 0$ for every $n < 0$?
- **(d)** Which ROC applies if $f[n]$ is two-sided?
- **(e)** Can the circle $|z| = 4$ lie in an ROC? Justify your answer.

4(c): When $f[n] = 0$ for $n < 0$

- Only a right-sided inverse can satisfy this condition
- This corresponds to the **exterior** ROC!

$$\boxed{|z| > 4}.$$



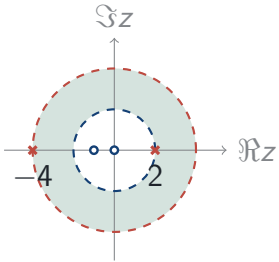
- The right-sided inverse indeed satisfies this condition:

$$f[n] = 2^n u[n] + (-4)^n u[n] = \left(2^n + (-4)^n\right) u[n].$$

4(d): When the signal is Two-sided

- Only **annular** ROCs are two-sided
- This expression has only one annular ROC

$$\boxed{2 < |z| < 4}.$$



- This corresponds to the following two sided inverse:

$$\boxed{f[n] = 2^n u[n] - (-4)^n u[-n-1].}$$

4(e): What happens for $r = 4$?

- No! $z = -4$ is a pole that lies on the circle with $r = 4$
- A circle in the ROC cannot pass through a pole!

$$F(z) = \frac{2z(z+1)}{(z-2)(z+4)}$$

